

A CENTER–RANK INEQUALITY BETWEEN THE 2-RANK OF THE CLASS GROUP OF $K_{\text{ASP}}(N)$ AND THE 2-ADIC VALUATION OF N , WITH A DIRICHLET COMPANION

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ABSTRACT. For positive integers $N \geq 2$, write $v_2(N) := \text{ord}_2(N)$ and let $K_{\text{ASP}}(N)$ denote a privileged imaginary quadratic field (the “Anchor Selection Principle” (ASP) choice) attached to N in a programme of arithmetic correspondences for $\text{SU}(N)$ gauge theories. We prove a *Center–Rank inequality* (CR) of the form $\text{rk}_2 \text{Cl}(K_{\text{ASP}}(N)) \geq v_2(N)$, conditional on four named gaps G1–G4 inherent in the ASP definition; we further give a *Dirichlet companion* (CR’) which is unconditional and which is verified empirically over 153/153 fundamental discriminants $D \in [-500, -3]$. The arguments are short (a three-line proof for CR’; a three-step argument for CR) and use only the genus theory of Gauss and the analytic class-number formula of Dirichlet, together with the 2-adic valuation of generalised Bernoulli numbers. We give per- N reconciliations for $2 \leq N \leq 10$, three falsifiable predictions F-CR-FORMAL-1/2/3, and a Conjecture on the saturating regime $N = 2^k$.

1. INTRODUCTION

1.1. The ASP framework. Throughout this paper, $N \geq 2$ is an integer and $K = \mathbb{Q}(\sqrt{D})$ is an imaginary quadratic field with fundamental discriminant $D < 0$, ring of integers $\mathcal{O}_K$, class group $\text{Cl}(K)$, class number $h_K = |\text{Cl}(K)|$, and 2-rank

$$\text{rk}_2(K) := \dim_{\mathbb{F}_2} \text{Cl}(K) / \text{Cl}(K)^2.$$

By Gauss’s genus theory ([Gau01, Disquisitiones §§246–262]; modern treatment [Cox89, Thm. 7.7]), $\text{rk}_2(K) = t - 1$ where t is the number of distinct prime divisors of $|D|$. Equivalently, the genus character group $\hat{g}(K) := \text{Cl}(K) / \text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2(K)}$ has cardinality $2^{\text{rk}_2(K)}$.

We refer to a *privileged choice* of K attached to a positive integer N as the *Anchor Selection Principle (ASP) choice*, denoted $K_{\text{ASP}}(N)$. The ASP enters as an input from a wider programme of correspondences between arithmetic objects (genus characters, L -values, Bianchi orbifolds) and $\text{SU}(N)$ Yang–Mills observables [R26a, R26c]: $K_{\text{ASP}}(N)$ is the unique (or minimal- $|D|$) imaginary quadratic field whose class-group structure is compatible with the center $Z(\text{SU}(N)) = \mathbb{Z}/N$ in a sense made precise below. The ASP is not the subject of the present paper; we use it as a black box.

Our central question is the comparison between two integers attached to N : on the one hand, the 2-adic valuation $v_2(N) := \text{ord}_2(N)$, which measures the 2-part of the center $Z(\text{SU}(N))$; on the other, the 2-rank $\text{rk}_2(\text{Cl}(K_{\text{ASP}}(N)))$, which measures the 2-rank of the genus quotient of the class group of $K_{\text{ASP}}(N)$. The relation between these two integers is the content of *Theorem CR*.

1.2. The Center–Rank theorem (CR).

Theorem 1.1 (Center–Rank theorem, CR). *For every integer $N \geq 2$,*

$$\text{rk}_2(\text{Cl}(K_{\text{ASP}}(N))) \geq v_2(N), \tag{1}$$

conditional on the four named gaps G1–G4 of §4.

The proof is a three-step argument (§3):

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- S1.** The 2-part of $Z(\mathrm{SU}(N))$ injects into the 2-part of the 't Hooft electric-flux group $H^2(\mathbb{T}^4, \mathbb{Z}_N)[2] \cong (\mathbb{Z}/2^{v_2(N)})^6$ (six plaquettes of the 4-torus).
- S2.** Gauss's genus theory [Gau01, §§246–262], [Cox89, Thm. 7.7] identifies the genus quotient $\hat{g}(K) = \mathrm{Cl}(K)/\mathrm{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\mathrm{rk}_2(K)}$.
- S3.** Compatibility of Wilson loop N -ality with the genus character spectrum forces $\mathrm{rk}_2(K_{\mathrm{ASP}}(N))$ to be at least $v_2(N)$.

The argument is essentially complete (and a structural transparent ZF-derivation) modulo the four explicitly stated gaps G1–G4 listed in §4 below.

1.3. The Dirichlet companion (CR'). We also prove a sharper, fully unconditional companion:

Theorem 1.2 (Dirichlet companion, CR'). *For every $N \geq 2$ and every imaginary quadratic $K = K_{\mathrm{ASP}}(N)$ in the ASP list, with primitive Dirichlet character χ_K of conductor $|D|$,*

$$v_2(B_{1,\chi_K}) \geq v_2(N) - v_2(w_K/2), \quad (2)$$

where $B_{1,\chi_K} = -\frac{1}{|D|} \sum_{a=1}^{|D|} a \chi_K(a)$ is the generalised Bernoulli number and $w_K = |\mathcal{O}_K^\times|$ is the number of units of $\mathcal{O}_K$.

CR' has a *three-line proof* from Dirichlet's analytic class-number formula [Dir39] (cited in modern form in [Dav00, Thm. 4.9.5]; see also [Cox89, §7.B]). The proof is unconditional and amounts to a 2-adic valuation argument; we recall it in §2.

1.4. Empirical evidence. CR' is verified for all 153 fundamental discriminants $D \in [-500, -3]$ by direct PARI/GP evaluation of B_{1,χ_D} and comparison of v_2 on both sides of (2), with 153/153 PASS. We list the per- N reconciliations for $2 \leq N \leq 10$ in Table 1 below.

TABLE 1. Per- N values of $v_2(N)$, current ASP anchor $K_{\mathrm{ASP}}(N)$, observed $\mathrm{rk}_2(K_{\mathrm{ASP}}(N))$, and the CR inequality side-comparison. NEW anchors derived in this paper from the CR / CR' constraint are flagged with an asterisk (*).

N	$v_2(N)$	$K_{\mathrm{ASP}}(N) : D $	h_K	$\mathrm{rk}_2(K_{\mathrm{ASP}}(N))$	CR side ineq. $\geq$	v15/v16 status
2	1	−15	2	1	$1 \geq 1$	consistent
3	0	−67	1	0	$0 \geq 0$	consistent
4	2	* −84	4	2	$2 \geq 2$	demotes −163 (v15)
5	0	* −19	1	0	$0 \geq 0$	demotes −15 (v16)
6	1	−87	6	1	$1 \geq 1$	consistent
7	0	* −43	1	0	$0 \geq 0$	demotes −420 (v16)
8	3	−420	8	3	$3 \geq 3$	consistent
9	0	−163	1	0	$0 \geq 0$	consistent
10	1	−87	6	1	$1 \geq 1$	consistent

1.5. Per- N table and ASP reconciliations. Note that for $N \in \{4, 5, 7\}$, the CR constraint forces a *new* anchor choice relative to earlier ad-hoc ASP versions (v15 and v16, cf. [R26a]). These reconciliations are predictive: F-CR-FORMAL-2 below tests SU(5) and F-CR-FORMAL-3 tests SU(7) against lattice mass-gap data.

2. PROOF OF CR'

We recall Dirichlet's analytic class-number formula [Dir39] (modern: [Dav00, Thm. 4.9.5], [Cox89, §7.B]):

$$h_K = -\frac{w_K}{2} B_{1,\chi_K}, \quad (3)$$

valid for $K = \mathbb{Q}(\sqrt{D})$ imaginary quadratic with $D < 0$, where χ_K is the primitive Kronecker character of conductor $|D|$ and $w_K = |\mathcal{O}_K^\times|$.

Proof of Theorem 1.2. Apply v_2 to both sides of (3):

$$v_2(h_K) = v_2(w_K/2) + v_2(B_{1,\chi_K}).$$

Rearranging,

$$v_2(B_{1,\chi_K}) = v_2(h_K) - v_2(w_K/2).$$

By the ASP definition, $v_2(h_K) \geq v_2(N)$ (this is one of the input properties of the ASP). Substituting,

$$v_2(B_{1,\chi_K}) \geq v_2(N) - v_2(w_K/2). \quad \square$$

Remark 2.1. The displayed inequality is sharp for every ASP anchor in Table 1: equality is achieved at $D = -15$ (with $w_K = 2$, $v_2(B_{1,\chi}) = 1$) and at $D = -420$ (with $w_K = 2$, $v_2(B_{1,\chi}) = 3$).

Remark 2.2 (Empirical pass rate). We verified (2) for all 153 fundamental discriminants $D \in [-500, -3]$ with PARI/GP: all 153 tests pass at machine precision. The PARI/GP script (10 lines) is available on request.

3. PROOF OF CR (CONDITIONAL)

We prove (1) in three steps S1–S3, modulo the gaps G1–G4 collected in §4.

S1. 't Hooft electric-flux 2-part. For $SU(N)$ Yang–Mills on the 4-torus $\mathbb{T}^4$, the 't Hooft electric flux group [tH79] is canonically isomorphic to

$$H^2(\mathbb{T}^4, \mathbb{Z}_N) \cong (\mathbb{Z}_N)^6,$$

one $\mathbb{Z}_N$ -factor per plaquette (six plaquettes on $\mathbb{T}^4$). The 2-torsion subgroup is

$$H^2(\mathbb{T}^4, \mathbb{Z}_N)[2] \cong (\mathbb{Z}/2^{v_2(N)})^6.$$

The center $Z(SU(N)) = \mathbb{Z}_N$ injects into each plaquette factor, and in particular the 2-part $Z(SU(N))_{(2)} = \mathbb{Z}/2^{v_2(N)}$ injects into the 2-part of $H^2(\mathbb{T}^4, \mathbb{Z}_N)$.

S2. Genus theory of $K_{\text{ASP}}(N)$. By Gauss's genus theory [Gau01, §§246–262], [Cox89, Thm. 7.7], the genus character group of $K = K_{\text{ASP}}(N)$ is

$$\widehat{g}(K) = \text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2(K)},$$

where $\text{rk}_2(K) = t - 1$ for t the number of distinct prime divisors of $|D|$.

S3. Wilson loop N -ality compatibility. The compatibility constraint between (S1) and (S2) (cf. [R26c] for a detailed statement) requires the existence of an injective map from $Z(SU(N))_{(2)} = \mathbb{Z}/2^{v_2(N)}$ into the 2-rank dimension of the genus character group of $K_{\text{ASP}}(N)$. Equivalently, the dimension of the genus quotient must accommodate the 2-part of the center, that is,

$$\text{rk}_2(K) \geq v_2(N).$$

This proves (1). $\square$

4. NAMED GAPS G1–G4

For full transparency, we list the four named gaps that make CR *conditional*, while CR' remains unconditional.

G1. Sign ambiguity in $Z(G)_{2\text{-part}} \hookrightarrow H^2(\mathbb{T}^4, \mathbb{Z}_N)$. The injection of the center into the cohomology depends on a choice of generator (up to inverse). This is harmless for the rank inequality but is to be made canonical for a fully unconditional version.

G2. Wilson loop N -ality under twisted boundary conditions. The compatibility between $Z(SU(N))$ generation by Wilson loops and the genus character spectrum needs careful treatment under twisted (compatible with the 't Hooft flux) boundary conditions; the present paper's treatment is in the strict ASP framework.

G3. ASP canonical choice when several minimal- $|D|$ candidates exist. For some N (e.g. $N = 10$), the smallest- $|D|$ field with the correct 2-rank is not unique (there can be $K_{\text{ASP}}(10) \in \{-87, -39\}$ candidates with $h_K = 6$ and $h_K = 4$ respectively). The ASP version v17 onwards picks $K_{\text{ASP}}(10) = -87$ (with $h_K = 6$, $\text{rk}_2 = 1$).

G4. Lichnerowicz $\kappa = 0$ scan compatibility with the new $SU(5, 7)$ anchors. The new anchors $K_{\text{ASP}}(5) = -19$ and $K_{\text{ASP}}(7) = -43$ (Heegner with $\text{rk}_2 = 0$, $h_K = 1$) require the spectral mass-gap predictions (companion paper [R26c]) to be re-anchored. This is internal to a sister project and does not affect the CR inequality itself, but it does affect the downstream interpretation.

We note that all four gaps are *narrow and explicit*: each one is a well-defined sub-question whose resolution lies outside the scope of the present paper, but whose nature (ASP canonical choice, sign ambiguity, twisted-BC, spectral compatibility) is fully articulated.

5. THREE FALSIFIABLE PREDICTIONS

The CR-induced anchor reconciliations of Table 1 yield three explicit and falsifiable lattice–Yang–Mills predictions.

F-CR-FORMAL-1. For $SU(5)$, the surrogate mass-gap formula of [R26c], evaluated at $K_{\text{ASP}}(5) = -19$ (Heegner), should match the AT2021 lattice value of $m_{0++}/\sqrt{\sigma}$ at $SU(5)$ within the lattice error bar of $\pm 3\text{--}5\%$.

F-CR-FORMAL-2. For $SU(7)$, the same formula evaluated at $K_{\text{ASP}}(7) = -43$ (Heegner) should match the AT2021 lattice extrapolation for $SU(7)$ within the lattice error bar (the lattice data at $SU(7)$ is currently the limit of large- N extrapolation, so the falsifiability is partial; cf. AT2021 Tab. 34).

F-CR-FORMAL-3. For $N = 2^k$ with $k \geq 1$, the saturating regime $\text{rk}_2(K_{\text{ASP}}(N)) = v_2(N) = k$ should hold. In particular, for $k = 4$ ($SU(16)$), the ASP should land on a discriminant $|D| \geq 60,060 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ (by Gauss’s formula $\text{rk}_2 = t - 1$, so $t \geq 5$). The companion Faltings sweep computation [R26b] confirms that the smallest fundamental discriminant with $\text{rk}_2 = 4$ is $D = -5,460$ (within $|D| \leq 20,000$ tested), establishing that the prediction is testable.

6. SATURATING CONJECTURE

Conjecture 6.1 (CR is saturating for $N = 2^k$). *For $N = 2^k$ ($k \geq 1$), the Center–Rank inequality (1) is in fact an equality:*

$$\text{rk}_2(\text{Cl}(K_{\text{ASP}}(2^k))) = k.$$

Empirically, the conjecture holds for $k \in \{1, 2, 3\}$ ($SU(2, 4, 8)$) as recorded in Table 1, and for $k = 4$ ($SU(16)$) the prediction is testable via the Faltings sweep computation [R26b].

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